

Plotting with LaTeX

This article is a tutorial on plotting graphs with LaTeX. It contains a number of examples that will be of immense use to readers who wish to try it out. It also has a comprehensive list of references, which will be of further use.



The LaTeX package for plotting is *pgfplots*. It's present in all full distributions of LaTeX, and permits the creation of consistent graphics with the typographic settings of the document, writing the instructions directly in the text source and ensuring the highest quality typical of LaTeX. In other words, *pgfplots* is a package that plots by coding. Further on in this article, an example of a flowchart will be shown. All the examples presented here are as standalone plots and are made with ProTeXt 3.1.5. Because the complete code is too long for a printed article, interested readers will find it freely available on <https://github.com/astoufe/latex>.

Data management

The first important thing to be aware of is how to use the data. Generally speaking, there are two main situations — a short data table and a long data table. The former is useful to put the table directly into the script, while in the case of the latter,

it's preferable to use an external file. For a short table, I use the *coordinates* of each point, as shown in the following code:

```
coordinates {
  (0.1,0.086)
  (...,...)
  (0.9,0.782)};
```

This data has been taken from Reference 1 given at the end of the article. Another way to use a short table is to use *table*, as shown in the following example:

```
table {
  x      y      z
0.324   0.213   0.463
...     ...     ...
0.150   0.700   0.150
};
```